

Task 2: Overfitting vs. Underfitting

1. Definition of Overfitting (High Variance)

Overfitting occurs when a machine learning model learns the training dataset too thoroughly, memorizing noise and random fluctuations rather than identifying the true underlying data patterns. Consequently, an overfitted model performs exceptionally well on the training data but fails to generalize to new, unseen test data.

2. Definition of Underfitting (High Bias)

Underfitting occurs when a machine learning model is too simple to capture the underlying structure and relationships within the data. As a result, an underfitted model performs poorly on both the training data and unseen test data.

3. Main Causes

Main Causes of Overfitting:

- Excessive model complexity (e.g., using high-degree polynomials or overly deep decision trees).
- Small training dataset relative to the number of parameters.
- High levels of noise or outliers in the training set.
- Lack of regularization constraints (such as L1/L2 penalty or dropout).

Main Causes of Underfitting:

- Model hypothesis is too simple (e.g., using a linear model for complex non-linear data).
- Insufficient or uninformative input features.
- Excessive regularization penalty that overly restricts model parameter values.

Under-training (stopping training iterations before the model